Registers

In Leiserson–Saxe form, each edge weight w is how many registers sit on that wire

The idea

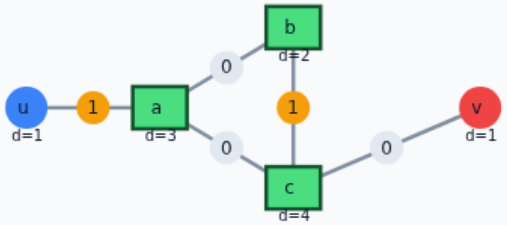
- E0 u→a has w=1; e2 b→c has w=1; zero-weight edges are combinational

Weights

REGISTER COUNTS

Weights

Step 1 / 5 · w · module01-03-register-count



Explanation

Edge w is the number of registers on that arc.

- w(e)

```
integers
```

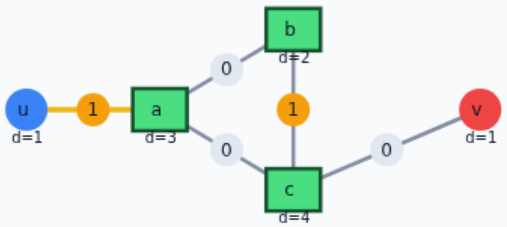
Weights

e0 has 1

REGISTER COUNTS

e0 has 1

Step 2 / 5 · e0 · module01-03-register-count



Explanation

u → a starts with one register (w=1).

- PI edge

```
w=1
```

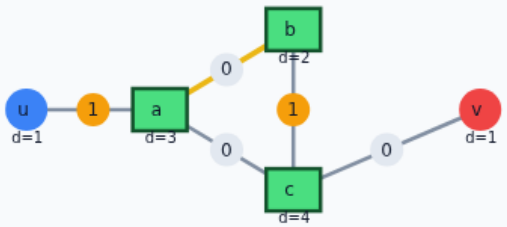
e0 has 1

e1 has 0

REGISTER COUNTS

e1 has 0

Step 3 / 5 · e1 · module01-03-register-count



Explanation

a → b is pure combo (w=0).

- Zero weight

```
path fuel
```

e1 has 0

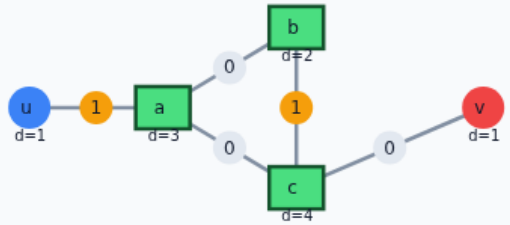
Sum = 2

REGISTER COUNTS

Sum = 2

Step 4 / 5 · sum · module01-03-register-count

regs=2



Explanation

Total registers = sum of w = 2 on the starter.

- area proxy

```
regs=2
```

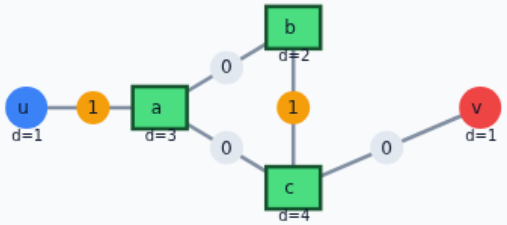
Sum = 2

Retiming moves w

REGISTER COUNTS

Retiming moves w

Step 5 / 5 · move · module01-03-register-count



Explanation

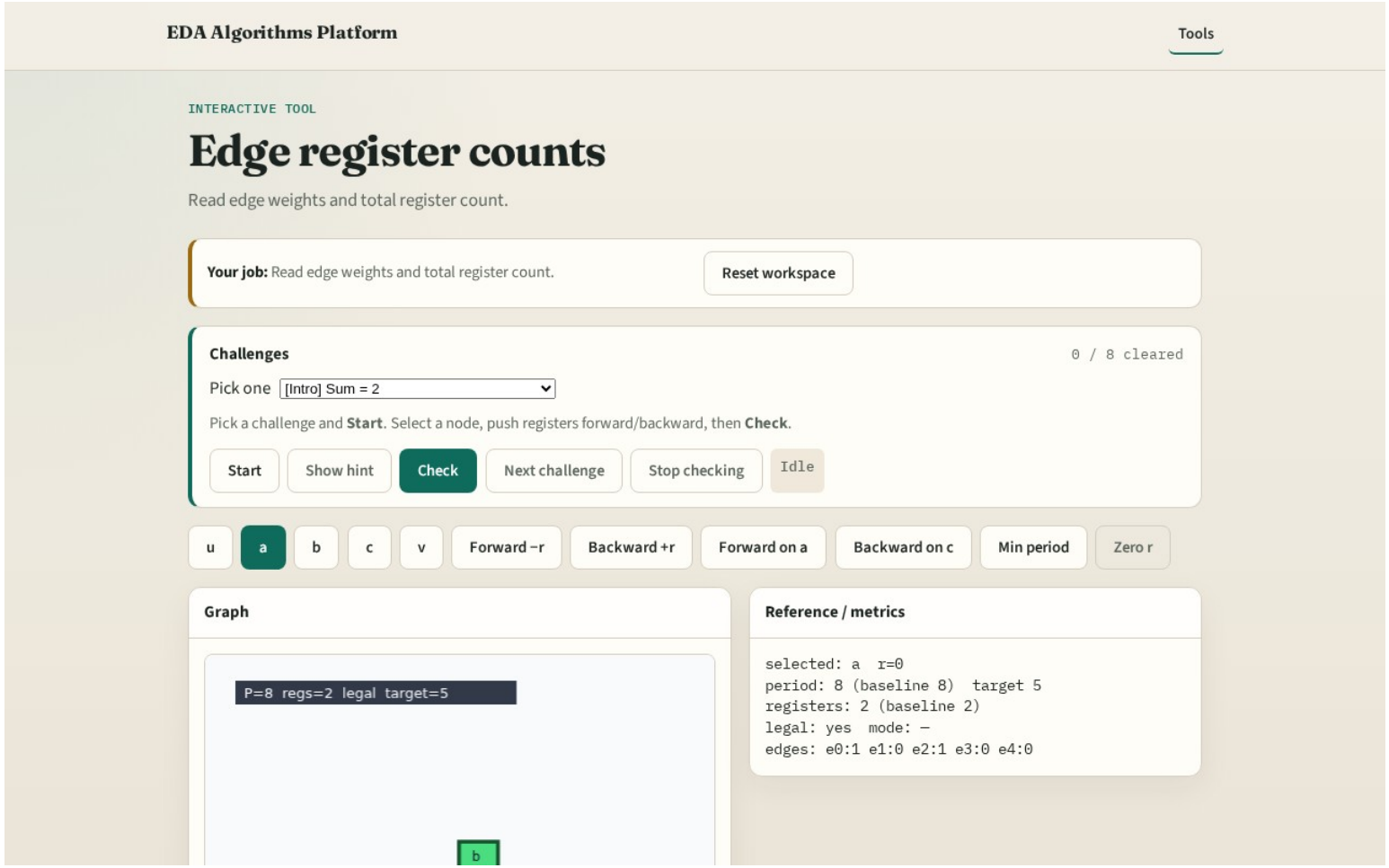
Changing r redistributes w without changing function (when legal).

- w_r

```
later labs
```

Retiming moves w

Browser lab track



Browser lab starter

Implement track

- Call total_registers(edges); assert sum == 2

Pitfalls

- Counting nodes instead of edges
- Forgetting w on u→a

Your turn

- Next: zero-weight path delay / clock period